

A Dual Classifier-Regressor Architecture for Heart Sound Onset/Offset Detection

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Abstract—Objective: Identifying the first and second heart sounds (S1 and S2) from phonocardiogram (PCG) signals is an essential step in automating the diagnosis of cardiac conditions such as irregular heartbeat, valve misfunctions, and heart failure. Recent research inspired by image segmentation has shown promise in utilising deep neural networks for point-wise PCG segmentation with the support of synchronised ECG. This paper shifts the focus from point-wise segmentation to identifying the onset/offset of S1 and S2 heart sounds in the PCG signal. **Methods:** We incorporate the electrocardiogram (ECG) signal and its keypoints to improve the detection of the heart sounds. Our proposed method employs a joint classifier-regressor architecture for predicting the probability and the location of transitions in the PCG signal. **Results:** When evaluated on the largest publicly available PhysioNet/CinC 2016 dataset, the proposed approach outperforms the existing state-of-the-art methods, achieving a sensitivity of 0.98 and a positive predictive value of 0.98 in identifying midpoints of S1 and S2 segments. It also identifies the onset/offset locations with an 11.11 ms error. **Conclusion:** It is evident that identifying the transitions simplifies, leading to better training and inference. **Significance:** In addition to achieving state-of-the-art results, this proposed approach could also be adapted to other physiological signals, such as respiration, blood pressure, or muscle activity, for locating regions of activity/interest.

Index Terms—Heart Sound Segmentation, Machine Learning, Phonocardiogram, Signal Processing

I. INTRODUCTION

Cardiovascular diseases (CVDs) were responsible for an estimated 17.9 million deaths globally in 2019, accounting for nearly one-third of all deaths worldwide [1]. This burden is projected to increase significantly through 2050, according to forecasts from the American Heart Association [2]. Early detection remains critical to reducing CVD-related mortality; however, traditional diagnostic methods, particularly auscultation, exhibit considerable variability in sensitivity and accuracy across clinicians [3], [4]. This inconsistency has driven the need for automated approaches to cardiac assessment [5]. n

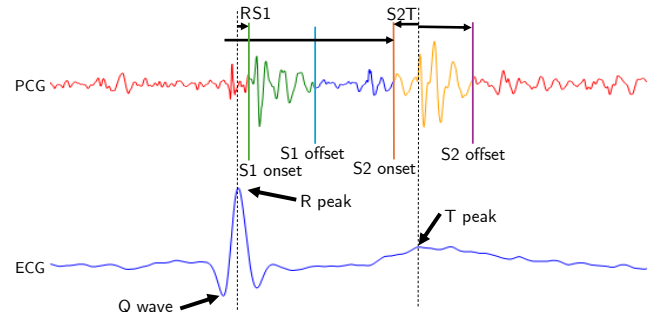


Fig. 1. The heart sound segments are characterised by their onset and offset locations and follow a specific order. The onset and offset locations of the heart sounds in the PCG signal are highly correlated with the PQRST-complex of the synchronous ECG signal, which can further aid in PCG segmentation.

The phonocardiogram (PCG), or heart sound, contains crucial information about the heart's mechanical function, particularly the performance of the heart valves, and is widely used in clinical practice to diagnose conditions such as arrhythmia, valve diseases, pulmonary hypertension, and heart failure [6]–[8]. As shown in Fig. 1, the heart sound is a complex signal composed of multiple segments: the first heart sound (S1), systole, second heart sound (S2), diastole, and occasionally murmurs and gallops indicative of specific heart conditions [8]. Accurate segmentation of the S1 and S2 components is a vital step in effective heart sound analysis and interpretation [9]–[11].

In addition to the phonocardiogram (PCG), the electrocardiogram (ECG) provides valuable information by capturing the

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heart's electrical activity. Given that electrical impulses initiate the myocardial contractions, which subsequently generate pressure changes that drive valve movements that produce the characteristic heart sounds, there exists a strong physiological correlation between ECG and PCG signals [12]. As illustrated in Fig. 1, the first heart sound (S1) occurs shortly after the R-peak (20 ± 5 ms [13]), whereas the second heart sound (S2) follows the end of the T-wave (24 ± 6 ms [13]). This relationship enables the use of ECG signal keypoints (PQRST-peaks) to enhance PCG-based heart sound segmentation. Furthermore, advancements in portable and miniaturized ECG sensors, as well as stethoscopes equipped with ECG recording capabilities, support the feasibility of ECG-PCG fusion in real-world clinical and wearable applications [14].

Early approaches to heart sound segmentation primarily relied on phonocardiogram (PCG) signals combined with classical signal processing techniques. Common methods included wavelet transforms [15], [16], Hilbert transforms [17], and envelope-based techniques such as Shannon energy envelopes [18]. Subsequently, model-based approaches gained prominence, offering a more structured representation by explicitly modeling heart sound sequences as state transitions. Probabilistic frameworks, such as hidden Markov models (HMMs) and hidden semi-Markov models (HSMMs), have been extensively utilized for this task. Notably, Springer et al. [19] proposed a logistic regression-HSMM that combined both probabilistic and machine learning approaches. The incorporation of auxiliary information from the ECG, such as R-peak onset and T-wave termination, into the HSMM framework has further improved segmentation accuracy [20]. However, despite their effectiveness in controlled environments, these methods often encounter challenges when applied to signals with high inter-patient variability, varying heart rates, or poor signal quality, particularly in pathological cases.

With the advent of deep learning based image segmentation, recent approaches have shifted towards data-driven models capable of learning temporal and spatial representations directly from the input signals. For instance, long short-term memory (LSTM) networks [21] and U-Net-based deep neural networks (DNNs) integrated with empirical wavelet transform (EWT) [22] have been applied for PCG-based segmentation. Renna et al. [6] used convolutional neural networks (CNNs) in conjunction with temporal modeling techniques such as HMM and HSMM to incorporate the flexibility of deep-learning methods with the predictability of temporal models. Additionally, convolutional LSTM architectures have also been explored for end-to-end heart sound segmentation [23].

Multimodal frameworks that fuse ECG and PCG signals have demonstrated improved robustness and accuracy, leveraging the physiological correlation between the two modalities. The use of ECG features such as R peaks and RT/RR intervals has been shown to assist in segmenting PCG signals by aligning cardiac events [24]. Moreover, Oliveira et al. [25] introduced a coupled hidden semi-Markov model (CHSMM) for state sequence identification, while Chen et al. [23] incorporated duration-specific parameters, such as estimated heart rate (eHR) and estimated systolic duration (eSys), into an LSTM-based framework, referred to as duration-LSTM. Silva

et al. [12] utilized the contextual information from preceding segments to enhance prediction in subsequent frames. R-peak detection was employed as a preprocessing step by Huang et al. [26] to identify individual cardiac cycles, which were mapped to a fixed window size. This enables the use of deep-learning models that are limited to a fixed input size.

A. Contributions

Many deep learning based methods rely on point-wise classification of the phonocardiogram (PCG) signal. While effective in some contexts, a key challenge faced in this approach is in capturing the unique temporal dynamics of heart sounds, which follow a fixed cyclic pattern (S1, systole, S2, diastole) with S1 and S2 durations in the range 90-130 ms [27]. Point-wise classification can lead to noisy predictions and overfitting due to its sensitivity to transient signal variations. To address these issues, existing methods frequently incorporate temporal models such as HSMMs as post-processing steps [6], [8] to enforce temporal consistency and physiological plausibility.

Instead of classifying each time point independently, we propose a different formulation of the problem: directly detecting the onset and offset of heart sound segments. This approach simplifies the task by focusing on the key transition points, inherently captures the temporal structure of the signal, and reduces sensitivity to noise, thereby offering a more robust alternative to traditional point-wise classification. This breaks down the segmentation task into two simpler sequential detection tasks: (1) determining whether a cardiac state transition occurs within a given signal window, and (2) localizing the temporal position of identified transitions.

In this paper, we present the following contributions:

- 1) We propose a novel approach to heart sound segmentation by identifying the onset and offset points of heart sound segments.
- 2) We develop a joint classifier-regressor architecture for localization of each transition on a given window, which predicts the existence and position of each transition.
- 3) We develop an adaptive stitching algorithm to combine the probability and location predictions for each window into a continuous segmentation output.
- 4) We conduct comprehensive experiments on a publicly available dataset and demonstrate that the proposed method achieves state-of-the-art results.

B. Paper Structure

The remainder of this article is organized as follows. Section II presents the proposed segmentation approach, detailing the key components of the overall pipeline. Section III describes the experimental setup, including dataset characteristics, preprocessing steps, and training procedures. In Section IV, we define the evaluation metrics and report the quantitative performance of our method in comparison with existing state-of-the-art approaches. Section V offers a critical analysis of the results, discusses the reliability of commonly used evaluation metrics in PCG segmentation, and outlines the limitations of our study. Finally, Section VI summarizes the key findings and suggests directions for future research.

II. METHOD

A. Concept

1) *Onset/Offset Detection*: Rather than performing point-wise classification across the entire cardiac cycle, our approach focuses on identifying key transition events; specifically, the onset and offset of the first (S1) and second (S2) heart sounds. This reformulation reduces the complexity of the task from classifying multiple points to detecting four major transitions per cycle. Each of these transitions corresponds to a physiologically meaningful event that typically induces a noticeable change in the PCG waveform, facilitating more accurate temporal localization. By targeting these discrete transitions, the model is explicitly trained to learn the precise timing of heart sounds. The loss function is designed based on the temporal error in predicting these transitions, rather than the misclassification of individual time samples. This strategy enhances temporal precision in segment boundary detection and improves the overall heart sound identification accuracy.

2) *ECG Integration*: Cardiac auscultation can be significantly enhanced by integrating synchronous ECG signals, leveraging the intrinsic link between the heart's electrical activity and its mechanical functions [12]. Electrical events such as depolarization and repolarization directly drive muscle contractions and cause valve movements, which in turn produce the sounds captured by a PCG. The ECG provides temporal markers for key cardiac events: the QRS complex corresponds to ventricular depolarization and atrial repolarization, while the T wave reflects ventricular repolarization and the onset of relaxation. These events exhibit well-established temporal relationships with PCG features. For instance, the Q wave typically precedes the first heart sound (S1) by 40–90 ms due to the rise in ventricular pressure that closes the atrioventricular valves [28]. Similarly, the T-wave peak aligns closely with the second heart sound (S2), which results from the closure of the semilunar valves during ventricular relaxation. Therefore, we extract the locations of R-peaks and T-peaks within the ECG and provide them along with the ECG signal as inputs to the proposed method. Empirically, we observe that this aids in detecting the existence of transitions and distinguishing between S1 and S2.

B. Problem Formulation

This work addresses the segmentation of the phonocardiogram (PCG) signal by framing it as a key-point detection problem within a fixed-length window (L_{window}). Let

$$w_i = (w_i^p, w_i^e, w_i^s)$$

denote the i -th window from the signals, where w_i^p , w_i^e , and w_i^s represent the PCG signal, the synchronized ECG signal, and the corresponding label sequence, respectively. The label sequence w_i^s encodes segment indices at each time point, corresponding to the four cardiac phases: S1 (index = 1), systole (index = 2), S2 (index = 3), and diastole (index = 4).

Conventional segmentation approaches attempt to directly estimate w_i^s given w_i^p and w_i^e using point-wise predictions. In contrast, we reformulate the problem to focus on detecting

cardiac state transitions within each window. Specifically, we define a target representation for each window i as,

$$t_i = (t_i^{\text{prob}}, t_i^{\text{loc}})$$

based on the presence and location of transition points within the i -th window. The four types of transitions considered are:

$$\begin{aligned} \text{tr}_1 &: \text{diastole} \rightarrow \text{S1}, & \text{tr}_2 &: \text{S1} \rightarrow \text{systole}, \\ \text{tr}_3 &: \text{systole} \rightarrow \text{S2}, & \text{tr}_4 &: \text{S2} \rightarrow \text{diastole}. \end{aligned}$$

The vector $t_i^{\text{prob}} \in [0, 1]^4$ indicates the presence of each transition type within the window. For ground-truth annotations, t_i^{prob} is binary-valued, while the predicted counterpart $\hat{t}_i^{\text{prob}}$ yields continuous scores in $[0, 1]$. The vector $t_i^{\text{loc}} \in [-1, L_{\text{window}} - 1]^4$ specifies the location (in sample indices) of each transition, with -1 indicating the absence of a transition within the window.

Consequently, the learning objective of the model f is,

$$f(w_i^p, w_i^e) \rightarrow (\hat{t}_i^{\text{prob}}, \hat{t}_i^{\text{loc}})$$

such that it minimizes the error between the predicted and true probabilities and the localization error of correct transitions.

Joint Classification-Regression Model

To accurately detect heart sound transition points in phonocardiogram (PCG) segmentation, we propose a joint classification-regression model that integrates information from both PCG and ECG signals. The model is designed to predict not only the existence of each transition type but also its temporal location within a fixed-length window. To leverage the temporal association between ECG morphology and PCG heart sounds, R-peaks and T-waves are first detected from the ECG signal. Their locations are encoded as binary indicator sequences and subsequently Gaussian-smoothed to provide soft temporal cues to the encoder regarding the proximity of R-peaks and T-waves.

As illustrated in Fig. 2, the model comprises two primary heads: a classification head that predicts the existence probabilities of the four transition types, and a regression head that estimates their respective locations. Empirical observations indicate that the classification head is particularly sensitive to the quality of the ECG signal, often degrading in performance when the ECG is corrupted by noise. To address this, we introduce an additional PCG-only classification pathway that operates independently of the ECG input, thereby enhancing robustness under noisy conditions.

During inference, for each window w_i of length L_{window} , the model outputs the following:

- **Regression Output** $\hat{t}_i^{\text{loc}} \in \mathbb{R}^4$: Estimated positions of each transition type within the window.
- **Classification Output (Fusion Model)** $\hat{t}_i^{\text{prob}_1} \in [0, 1]^4$: Transition existence probabilities based on both PCG and ECG inputs.
- **Classification Output (PCG-Only Model)** $\hat{t}_i^{\text{prob}_2} \in [0, 1]^4$: Transition existence probabilities estimated using only PCG input.

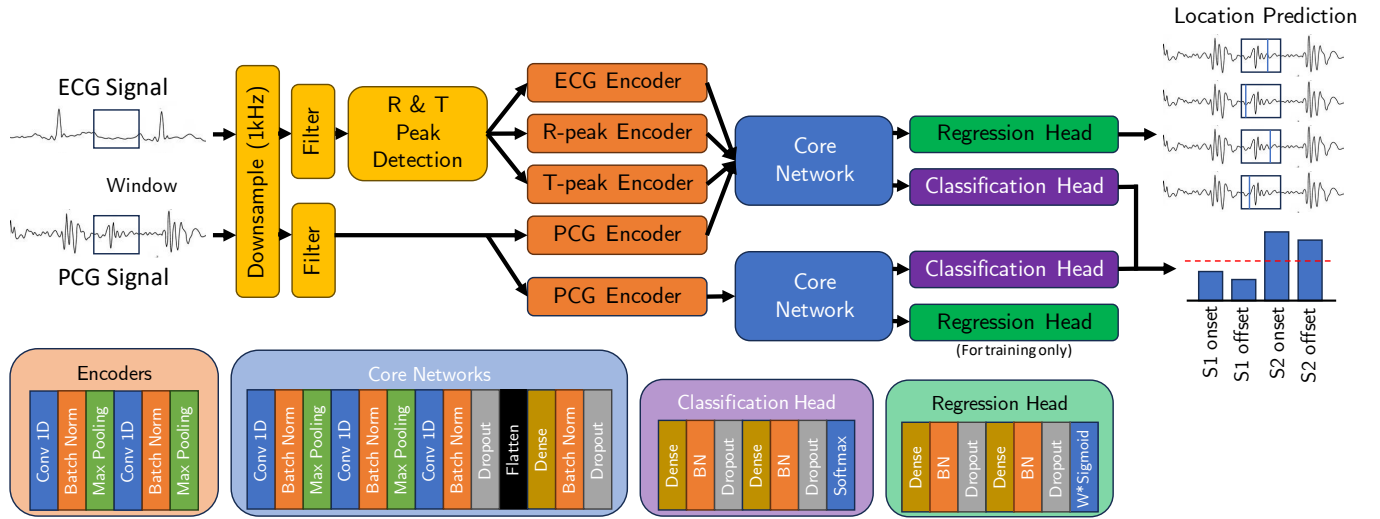


Fig. 2. Overview of the proposed joint classification-regression model architecture. One window of each PCG and ECG signal is fed to the corresponding encoder. The model predicts the likelihood of the existence of each transition and the position of the corresponding transition if it exists.

The final transition existence probability vector $\hat{t}_i^{\text{prob}} \in [0, 1]^4$ is computed by averaging the two classification outputs:

$$\hat{t}_i^{\text{prob}} = \frac{\hat{t}_i^{\text{prob}_1} + \hat{t}_i^{\text{prob}_2}}{2}.$$

The model is trained using a composite loss function $\mathcal{L}$, defined as a weighted sum of a regression loss $\mathcal{L}_{\text{reg}}$ and a classification loss $\mathcal{L}_{\text{cls}}$:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{cls}} + (1 - \alpha) \cdot \mathcal{L}_{\text{reg}}, \quad (1)$$

where $\alpha \in [0, 1]$ is a tunable hyperparameter controlling the balance between classification and regression objectives.

The **regression loss** $\mathcal{L}_{\text{reg}}$ is defined as the Mean Squared Error (MSE) between the predicted locations $\hat{t}_i^{\text{loc}}$ and the ground truth positions t_i^{loc} . For each transition type, the loss is computed only if the transition exists within the window (i.e., the corresponding entry in t_i^{loc} is not -1).

The **classification loss** $\mathcal{L}_{\text{cls}}$ is computed as the sum of weighted binary cross-entropy (BCE) losses across the four transition types, with increased weights assigned to the positive class to compensate for class imbalance due to the relative sparsity of transitions.

The weighted BCE for a single transition type is given by:

$$\text{BCE}(t, \hat{t}) = -\beta \cdot t \cdot \log(\hat{t}) - (1 - \beta) \cdot (1 - t) \cdot \log(1 - \hat{t}), \quad (2)$$

where $t \in \{0, 1\}$ is the ground truth label, $\hat{t} \in [0, 1]$ is the predicted probability, and β is the positive class weight.

During training, an additional regression head is attached to the PCG-only model to enhance the learning of the shared encoder and core network. This auxiliary head improves feature representation by providing location supervision specific to the PCG signal. However, it is omitted during inference, as the location predictions from the fusion model are considered sufficient and render the PCG-only regression output redundant.

Stitching Algorithm

To make predictions over long PCG-ECG recordings, the model processes the input signals using overlapping windows. As each segment of the signal is analyzed multiple times due to window overlap, a stitching algorithm is required to combine these multiple predictions and produce a coherent segmentation mask aligned with the original signal.

We propose an adaptive stitching algorithm that leverages both statistical consensus and physiological constraints to improve transition detection reliability. This method enforces the canonical cyclic order of heart sound states, $S1 \rightarrow S2 \rightarrow \text{diastole}$, and incorporates a minimum interval (> 50 ms) between consecutive transitions of the same type to reflect realistic cardiac physiology.

Windowing and Buffering: The signal is segmented using a sliding window of length N and stride s . Each signal bin of length s is therefore evaluated multiple times, with each evaluation contributing a predicted transition type, its temporal position (relative to the window), and its associated confidence (probability).

For each transition type $j \in \{1, 2, 3, 4\}$, the algorithm maintains:

- A buffer $D_{\text{pos}}[j]$ to accumulate detected transition positions within the current context.
- A buffer $D_{\text{prob}}[j]$ to store the corresponding prediction probabilities.

Key Mechanisms for Physiological Consistency: Three mechanisms are used to enforce physiologically plausible detection:

- 1) **Consensus-Based Confirmation:** A transition of type j is confirmed only if at least n_t predictions in $D_{\text{pos}}[j]$ exceed the adaptive threshold. The final position of the confirmed transition is computed as the mean of these n_t positions to reduce sensitivity to noise and jitter.
- 2) **Freezing Period (τ_f):** After confirming a transition of type j , a refractory period τ_f is enforced during which further detections of the same type are ignored. This prevents over-segmentation and redundant detections.

Algorithm 1 Adaptive Transition Stitching Algorithm

Require: Overlapping predictions $\{(\hat{t}_i^{\text{loc}}, \hat{t}_i^{\text{prob}})\}$, stride s , window length N , freezing time τ_f , vote threshold n_t

Ensure: Final segmentation mask for the full signal

- 1: Initialize buffers $D_{\text{pos}}[j] \leftarrow \emptyset$, $D_{\text{prob}}[j] \leftarrow \emptyset$ for each transition type $j \in \{1, 2, 3, 4\}$
- 2: Initialize last confirmed time $\text{last_confirm}[j] \leftarrow -\infty$ for all j
- 3: **for** each window i **do**
- 4: **for** each transition type j **do**
- 5: $p \leftarrow \hat{t}_i^{\text{loc}}[j]$
- 6: $q \leftarrow \hat{t}_i^{\text{prob}}[j]$
- 7: $t_{\text{abs}} \leftarrow i \cdot s + p$
- 8: **if** $p \geq 0$ **and** $q \geq \text{AdaptiveThreshold}(j)$ **and** $t_{\text{abs}} - \text{last_confirm}[j] \geq \tau_f$ **then**
- 9: Append p to $D_{\text{pos}}[j]$
- 10: Append q to $D_{\text{prob}}[j]$
- 11: **end if**
- 12: **if** $|D_{\text{pos}}[j]| \geq n_t$ **then**
- 13: Confirm transition of type j
- 14: $t_j^{\text{final}} \leftarrow \text{mean}(D_{\text{pos}}[j])$
- 15: Update $\text{last_confirm}[j] \leftarrow t_{\text{abs}}$
- 16: Clear $D_{\text{pos}}[j]$, $D_{\text{prob}}[j]$
- 17: Update segmentation mask using t_j^{final}
- 18: **end if**
- 19: **end for**
- 20: **end for**
- 21: Aggregate per-bin segmentation masks
- 22: Discard first and last 350 ms from final mask

- 3) **Adaptive Thresholding:** The probability threshold for accepting a transition is dynamically lowered if the transition is expected next in the canonical sequence, given the most recently confirmed transition. This prioritizes transitions that maintain a valid sequence.

The confirmed transitions are used to construct a segmentation mask for each bin, which is subsequently aggregated to reconstruct the full-length segmentation.

III. EXPERIMENTS

This section describes the experimental setup, including the dataset, preprocessing steps, and implementation details.

A. The Dataset

We conduct our experiments using the Massachusetts Institute of Technology Heart Sounds Database (MITHSDB), part of the publicly available PhysioNet/CinC 2016 Challenge training set [29]. MITHSDB contains synchronized electrocardiogram (ECG) and phonocardiogram (PCG) recordings from 121 subjects under a variety of clinical and non-clinical conditions, capturing realistic environments with varying noise levels. The dataset includes normal heart sounds as well as pathological conditions such as mitral valve prolapse, benign murmurs, and aortic diseases, ensuring broad coverage; however, specific pathological details of each recording have not been provided in the dataset.

Each recording had been manually corrected following initial automated annotation using Springer's algorithm [19]. The full dataset comprises 409 annotated recordings, ranging between 9-37 seconds in duration. Of these, 29 recordings were labeled as noisy, and 4 had missing modalities, resulting in 376 recordings being used for our experiments. The signals have been recorded at 2 kHz. The heart sounds were collected from nine different locations, including the aortic area, pulmonary area, tricuspid area, and mitral area.

B. Preprocessing

ECG and PCG signals are first resampled to 1 kHz. A fifth-order band-pass Butterworth filter is then applied: for PCG, the passband is 25–400 Hz, and for ECG, the passband is 1–45 Hz, effectively removing frequency components outside their respective physiological ranges.

Following filtering, we normalize each signal window. However, instead of using the absolute maximum value in the signal window, which is highly sensitive to noise, we use the 98th percentile of the absolute maximum to perform the normalization, which is more robust against outliers and noise. R- and T-peak locations are automatically extracted using the NeuroKit2 library [30], which identifies QRS complexes via signal gradient steepness, with R-peaks detected as local maxima. These discrete peaks are then convolved with a Gaussian kernel to produce smooth event markers.

C. Implementation Details

We set the window size (L_{window}) to be higher than the possible S1 and S2 region lengths (~ 200 ms). We empirically found that a window size of 400 ms with a 10 ms stride showed good performance on the validation data. To address class imbalance in transition detection, the positive class weight for the cross-entropy loss is set to $\beta = 1.5$, reflecting the lower frequency of transition events relative to non-transition windows. The overall loss function comprises classification and regression terms, with a weighting factor of $\alpha = 0.99$ to balance their contributions.

Following the protocol in [12], we adopt a 10-fold cross-validation scheme. For each fold, 10% of the data is reserved for testing, while the remaining 90% is further randomly split into 90% training and 10% validation subsets.

The model is implemented in PyTorch [31] and trained using mini-batch gradient descent with the Adam optimizer [32] ($\beta_1 = 0.9$, $\beta_2 = 0.99$), a batch size of 2048, and an initial learning rate of 5×10^{-4} . Early stopping is applied with a patience of four epochs based on validation loss to prevent overfitting. Training was performed on an NVIDIA GeForce RTX A6000 GPU, with total training time across all 10 folds approximately 2.5 hours.

IV. RESULTS

This section outlines the evaluation metrics used to assess the performance of our segmentation framework and compares it against previous state-of-the-art methods.

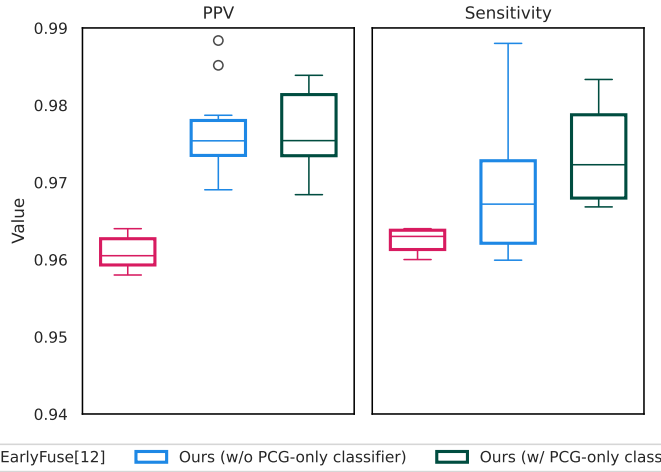


Fig. 3. Positive predictive value (PPV) and sensitivity obtained via 10-fold cross-validation for our model are shown. Results on Early-fuse [12] were not reported and therefore we estimated from the published plots.

A. Metrics

We evaluate the model's ability to recover the ground-truth state sequence of the PCG signal. To avoid edge effects from the prediction stitching algorithm, the first and last 350 ms of each recording are excluded from evaluation.

Midpoint Segmentation Accuracy: This is the widely used metric for evaluating heart sound segmentation since 2008 [33]. It compares predicted and ground-truth midpoints of the S1 and S2 sounds within a tolerance window of 60 ms. A predicted midpoint is considered a true positive (TP) if it lies within this tolerance window of a ground-truth midpoint of the same label. The sensitivity (Se) and positive predictive value (PPV) are then computed as:

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{PPV} = \frac{TP}{TP + FP} \quad (3)$$

where TP , FP , and FN denote true positives, false positives, and false negatives, respectively. These reflect how well the model identifies the occurrence of S1 and S2 events.

Classification Confusion Metrics: Since the model predicts the presence or absence of each transition type within a given window, classification performance must be evaluated for each transition independently. This is particularly important because multiple transitions may occur within the same window at elevated heart rates. We therefore report confusion matrices separately for each transition type to quantify both correct detections and misclassifications. In addition, we compute standard classification metrics, including precision, recall, and F1-score, to further assess the reliability of transition presence predictions.

Onset and Offset Time-Shift Errors: To assess the temporal precision of a segmentation, we propose evaluating the offset between predicted and ground-truth transition points. This measures the average absolute timing error across the four key cardiac transitions, diastole→S1, S1→systole, systole→S2, and S2→diastole, for correctly detected segments as:

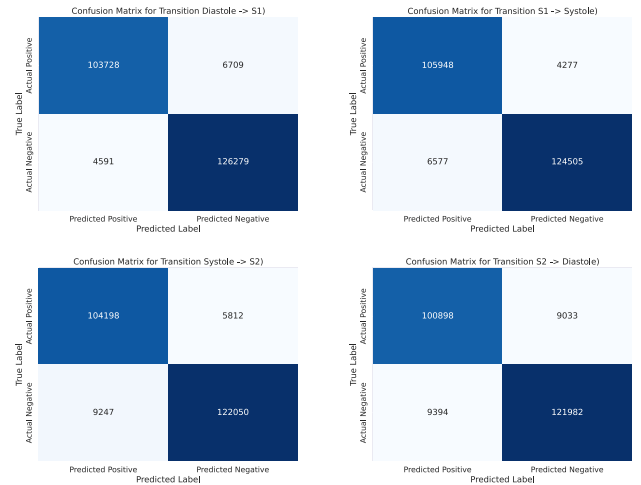


Fig. 4. Confusion matrices illustrating the performance of the multi-label transition detection task. Since a single window can include multiple transitions, each class is evaluated independently.

$$\Delta t_{\text{on},i} = t_{\text{on},i}^p - t_{\text{on},i}^g, \Delta t_{\text{off},i} = t_{\text{off},i}^p - t_{\text{off},i}^g \quad (4)$$

We report the mean and standard deviation of the absolute timeshift errors to capture the accuracy and variation in localizing the transition points. However, this metric is limited to windows in which both the ground truth and the prediction label the window as containing the same type of transition. Therefore, this needs to be evaluated alongside the classification accuracy.

Evaluation Summary: Our method is evaluated using a combination of traditional midpoint segmentation metrics (sensitivity and PPV), classification-based metrics for transition presence (confusion matrices, precision, recall, and F1-score), and the proposed onset/offset time-shift error. Together, these measures provide a comprehensive assessment of detection accuracy, classification reliability, and temporal alignment. This is particularly important in noisy or pathological recordings, where precise boundary localization and accurate identification of cardiac state transitions are clinically meaningful and essential for reliable downstream analysis.

B. Quantitative Results

We compare our method against two state-of-the-art approaches for PCG segmentation that incorporate both PCG and ECG modalities. Silva et al. [12] employ a U-Net architecture with modality-specific encoders for PCG and ECG, which are fused prior to classification. The model performs point-wise segmentation by predicting class probabilities for each time step. Huang et al. [26] adopt a two-stage approach: cardiac cycles are first segmented based on ECG R-peaks, followed by intra-cycle classification using a U-Net.

Midpoint Segmentation Accuracy: As summarized in Table I, our proposed approach demonstrates superior performance compared to both baseline models, in both sensitivity (Se) and PPV. These results suggest that our method achieves more accurate detection and a lower rate of false positives.

Notably, our final model consistently outperforms the method by Silva *et al.* [12] in segmentation accuracy across all 10 folds of cross-validation, as depicted in Fig. 3.

Although the original implementation and random seeds from [12] were not publicly available, which prevented exact replication of their cross-validation folds, the consistent improvements observed across all folds indicate the robustness and generalizability of our approach. However, Huang *et al.* [26] did not report cross-validation results, thereby limiting the scope of direct quantitative comparison with their method.

To further assess the contribution of the PCG-only classifier, we compared an earlier version of our model (excluding this component) with the final version, focusing on sensitivity and PPV. Both metrics were found to be normally distributed. A paired t-test revealed that the improvement in sensitivity is statistically significant ($p = 7 \times 10^{-4}$), while the difference in PPV is not statistically significant ($p = 0.062$). These findings indicate that the inclusion of the PCG-only classifier enhances the model's ability to correctly identify actual segments, reducing the likelihood of missing heart sound segments.

Due to the unavailability of per-fold results from [12], we could not conduct statistical comparisons. However, visual inspection of Fig. 3 shows that the distribution of metrics from our final model does not overlap with those from [12], suggesting a potentially significant improvement over the current state-of-the-art.

TABLE I

MIDPOINT SEGMENTATION ACCURACY ON PHYSIONET 16 DATASET.
PPV = POSITIVE PREDICTIVE VALUE, Se = SENSITIVITY

	PPV	Se
DL U-net [26]	96.3	96.55
Early-Fuse [12]	96.1	96.2
Ours (w/o PCG-only classifier)	97.5	96.7
Ours (w/ PCG-only classifier)	97.7	97.4

Classification Confusion Metrics: Fig. 4 presents the confusion matrices for the four cardiac state transitions predicted by the classification head of the proposed model. Across all transition types, the model correctly identifies the majority of analysis windows, indicating robust transition presence detection. Residual misclassifications, primarily arising from noise and window boundary effects, are effectively mitigated by the subsequent adaptive stitching algorithm, resulting in an overall sensitivity and positive predictive value approaching 98%. The strong classification performance is further quantified by the precision, recall, and F1-scores reported in Table II.

Onset and Offset Time-Shift Errors: Table III presents the onset and offset time-shift errors of our model. Although comparable timing metrics are not available from the baseline methods, our results demonstrate strong temporal localization, with average errors in the range of 10–13 ms. These are significantly smaller than both typical heart sound durations and the 60 ms tolerance used in midpoint segmentation metrics, suggesting clinically relevant precise boundary detection.

V. DISCUSSION

TABLE II

CLASSIFICATION PERFORMANCE FOR CARDIAC STATE TRANSITIONS.

Transition	Precision	Recall	F1-score
Diastole $\rightarrow$ S1	0.958	0.939	0.948
S1 $\rightarrow$ Systole	0.942	0.961	0.951
Systole $\rightarrow$ S2	0.918	0.947	0.933
S2 $\rightarrow$ Diastole	0.915	0.918	0.916

TABLE III

TIME SHIFT METRICS OBTAINED FOR THE CORRECTLY PREDICTED STATES.

	Dias $\rightarrow$ S1	S1 $\rightarrow$ Sys	Sys $\rightarrow$ S2	S2 $\rightarrow$ Dias
Mean Absolute Error	10.70	10.86	12.07	13.31
Standard Dev.	10.06	10.61	11.85	12.57

A. Implications from the Results

Our results consistently outperform existing methods across 10-fold cross-validation, demonstrating that reformulating PCG segmentation as a key-point detection task, rather than a point-wise classification problem, leads to smoother and more stable learning. This formulation not only improves detection accuracy but also yields low transition location errors, making the predicted transition points viable for downstream applications. These include tasks such as cardiac cycle-specific feature extraction, heart sound quality assessment, or pathological murmur localization.

Furthermore, the proposed approach is generalizable to other biosignals such as respiratory sounds, blood pressure waveforms, electroencephalograms (EEG), and electromyograms (EMG), where temporal segmentation and activity onset/offset detection are essential for further analysis or classification.

B. Metrics Used for Evaluating PCG Segmentation

The most commonly used metrics for evaluating PCG segmentation are Positive Predictive Value (PPV) and sensitivity based on the detection of segment midpoints. These were introduced by Schmidt *et al.* [33], although their work does not clarify the rationale behind focusing on midpoints, the relevance of these points to physiological interpretation, or the selection of the tolerance threshold (60 ms).

While these metrics effectively capture segmentation classification performance, they are becoming less discriminative in the context of modern models that routinely achieve PPV and sensitivity scores above 0.95. Additionally, reliance on midpoints may obscure certain types of errors. For instance, if the predicted onset and offset of a segment are misaligned, but symmetrically around the midpoint, the resulting midpoints can still appear close to the ground truth, even though the actual segment is misrepresented in duration and boundary alignment. As illustrated in Fig. 5, such time-shift errors can go undetected when using midpoint-based evaluation alone.

To address this limitation, we propose a complementary metric that explicitly evaluates the localization accuracy of transition boundaries. Specifically, we introduce the onset and offset time-shift error, as described in Section IV. This is computed as the average absolute temporal difference between predicted and ground-truth transition points.

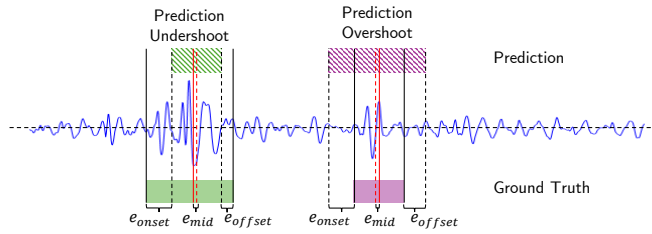


Fig. 5. Two examples demonstrating limitations of current midpoint-based metrics. Despite low error at the midpoints (e_{mid}) for S1 and S2, the onset and offset predictions exhibit substantial temporal misalignment (e_{onset} and e_{offset}).

We acknowledge that this metric cannot be calculated for unmatched transitions, i.e., when a ground-truth transition is missed or when a predicted transition does not correspond to any ground truth. Nonetheless, when used alongside PPV and sensitivity, this metric provides a more comprehensive assessment of segmentation performance. In particular, models that achieve both high PPV and sensitivity with low time-shift error can be considered more reliable for downstream tasks.

C. Limitations of the Proposed Approach

While the proposed method demonstrates strong performance on the given dataset, several limitations remain. One significant challenge arises when the ECG signal is noisy or unreliable. High inter-sample variability between clean and corrupted ECG recordings can degrade the quality of extracted features. In particular, inaccuracies in R- and T-peak detection performed using NeuroKit2 may propagate to downstream model inputs, leading to substantial prediction errors. Although our consensus-based stitching algorithm mitigates some of this degradation, improving robustness to ECG noise remains an important area for future research.

Another limitation pertains to the reduced accuracy of transition predictions when these occur near the boundaries of a window. This is a common issue in window-based segmentation frameworks, where contextual information near the edges is limited. Our stitching mechanism helps smooth such inconsistencies across overlapping windows, but more sophisticated approaches, such as attention-based boundary refinement, may result in lower localization error.

Moreover, the current formulation is restricted to detecting only S1 and S2 heart sounds. It does not account for the presence of additional cardiac events such as S3, S4, or pathological murmurs, which may be misclassified as S1 or S2. Although this limitation exists in all prior methods, it hinders clinical applicability, particularly in detecting disease-related anomalies. Addressing this gap will require access to more comprehensive datasets with annotated abnormal heart sounds, as well as model architectures capable of handling extreme class imbalance. While our model is already designed to manage the imbalance between transition presence and absence within a window, rarer classes such as S3, S4, and murmurs present a far more severe imbalance scenario.

Finally, the current model comprises approximately 1.5 million parameters, which, although comparable to existing

methods, poses a challenge for deployment on resource-constrained devices in real-time applications. Techniques such as model pruning, quantization, and knowledge distillation offer promising directions for compression. However, we are also investigating task-specific strategies that exploit domain knowledge, such as the physiological coupling between ECG and PCG, to simplify model design. Additionally, incorporating complementary signals such as photoplethysmography (PPG) or forcecardiography (FCG) could enrich the feature space, potentially enabling smaller and more efficient models without sacrificing performance.

VI. CONCLUSION

In this work, we introduced a novel heart sound segmentation framework that reformulates the task as onset/offset detection using a joint classification–regression model trained on synchronized PCG and ECG signals. Unlike traditional point-wise approaches, our method emphasizes precise localization of segment boundaries, leading to improved detection accuracy and temporal alignment. Experimental results on a real-world dataset demonstrate the effectiveness of this approach over existing state-of-the-art methods. The proposed framework is generalizable and opens opportunities for integrating other signals, such as the forcecardiogram (FCG), for heart sound segmentation and extension to other biosignals, such as respiration and blood pressure.

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